

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Minor

M.C.A. (Monsoon Semester-2023)

Course Name: Computer Architecture

Maximum Marks: 30

Course Code: CS 405

Total Time: 1 hours

Q1. a) What is the difference between Von Neumann Architecture and Harvard Architecture?. What are the different types of addressing modes that has evolved in Computer Systems?.

b) A 32 bit floating point binary number has 4 bit plus sign for exponent and 26 plus sign for mantissa. The mantissa is a normalized function. Numbers in mantissa and exponent are in a signed-magnitude representation. What are the largest and smallest positive quantities that can be represented excluding zero?.

(10)

Q2. a) What is the difference between RISC and CISC type of Computer Architecture?. Explain with examples.

b) Which one is better and Why?. How can one compensate the drawbacks of one over the other type of architecture.

(10)

Q3. a) What is T- state, Machine Cycle and Instruction Cycle in a Computer System?.

b) Explain the concept of pipelining technique in a computer Architecture system?.

(10)

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Mid-Term Exam
M.CA. (Winter Semester-2023)

Course Name: Computer Architecture
Course Code: CS 106

Maximum Marks: 40
Total Time: 2.0 hours

Q1. What is the difference between RISC and CISC type of Computer Architecture? Three enhancements with following speedups are proposed for a new computer:

Speedup₁=25; Speedup₂=15; Speedup₃=10; only one enhancement is useable at time

- (a) If enhancement 1 and 2 are each usable for 22% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 15?
- (b) Assume the enhancement can be used 20%, 35% and 15% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?

(3+7)

Q2. (a) What are the major performance parameters in designing computer architecture?

b) What are various types of addressing modes in Computer Architecture?

- (c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.0 and ii) reduce CPI instruction type B to 1.5. Which one is better?.

Type	CPI	Frequency
A	1.2	50%
B	2.4	30%
C	2.1	20%

(3+2+5
)

Q3. (a) What is the difference between SRAM and DRAM memory cell? Where are they used in the memory hierarchy system and why?

(b) What conclusion can one draw from the following observations in terms of computing performance

Type	CPU A (in Sec.)	CPU B (in Sec.)	CPU C (in Sec.)
Program A	60	40	30
Program B	510	220	90
Total Time	570	260	120

Q4. (a) What is the difference between Access time and Latency in a memory system? What is the difference Associative Mapping N Set Associative Mapping?

(b) Consider a two level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.

Calculate the average access time for the following parameters:

- i) Clock cycle is 1 ns
- ii) Miss cycle is 10 clock cycle
- iii) 2 percent of instruction is not found in Instruction cache
- iv) 9 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
Final Exam

M.CA. (Winter Semester-2023)

Shreya Pal

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 106

Total Time: 3.0 hours

Q1. (a) What is the difference between X86 and ARM type of Computer Architecture?

(b) Explain the concept of parallelism in Computer System.

(c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.2 and ii) reduce CPI instruction type B to 1.5. Which one is better?

Type	CPI	Frequency
A	1.5	50%
B	2.5	30%
C	2.1	20%

(3+2+5)

Q2. (a) What is flash memory and which part of memory hierarchy it is placed? Explain the term locality of reference.

(b) What conclusion can one draw from the following observations in terms of computing performance

Type	CPU A (in Sec.)	CPU B (in Sec.)	CPU C (in Sec.)
Program A	60	40	30
Program B	510	220	90
Total Time	570	260	120

(4+6)

Q3. (a) What is a Vector Processor?

(b) The great advantage of pipelining is the performance improvement with a shorter clock time.

Element	Register Clk to q	Register Set Up	Mux.	ALU	Mem. Read	Mem. Write	Register File Read	Register File Set Up
Parameter	t clk-to-q	t Setup	t Mux.	t ALU	t MEM read	t MEM write	t RF Read	t RF setup
Delay (ps)	30	20	25	200	250	200	150	5

i) What was the clock time and frequency of a single cycle CPU?

ii) What is the clock time and frequency of a pipelined CPU?

iii) What is the speed-up?

(4+6)

Q4. (a) What are the guiding parameters for defining the memory Hierarchy?

(b) Consider two different types of memory hierarchy. In the first case, consider a three level memory hierarchy with separate instruction and data cache in level 1 while the main memory is at level 2. In the second case, consider a three level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.

Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 2 ns
- ii) Miss cycle is 10 clock cycle
- iii) 2 percent of instruction is not found in Instruction cache
- iv) 9 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(2+8)

Q5. (a) What is the difference between Deep and Shallow pipelining technique?

(b) For the following reservation table

- (i) What are the forbidden latencies
- (ii) Show the state transition diagram
- (iii) List all the simple and greedy cycles
- (iv) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\tau=20\text{ns}$

	1	2	3	4	5	6
S1	X		X	X	X	
S2		X		X		X
S3		X	X		X	

	1	2	3	4	5
S1	Y		Y	Y	Y
S2		Y		Y	
S3	Y	Y			

(2+8)

Q6. (a) What are the different ways for cache optimization techniques.

(b) Suppose that in 1000 memory references there are 60 misses in L-1 cache are 30 misses in L-2 cache and cache are 10 misses in L-3 cache. What are various miss rates?

Assume that Miss Penalty $L-2$ is 180 clock cycles, Hit time $L-1$ is 1 clock cycle, Hit time $L-2$ is 6 clock cycle, Miss Penalty $L-3$ is 90 clock cycles, Hit time $L-3$ is 12 clock cycle

What will be the average memory access Time?. Ignore impact of write.

(4+6)